Theorie
das sollte man können
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1 The rank theorem

Let

$$T : V \to W$$

be a linear map, where V is finite dim. Then

$$\dim(\ker(T)) + \dim(\operatorname{Im}(T)) = \dim(V)$$

Proof.

Write $n := \dim(V)$.
Let $u_1, ..., u_k$ be a basis for $\ker(T)$

$$\operatorname{span}(u_1, ..., u_k) = \ker(T)$$

Extend the basis of $\ker(T)$ to a basis for V

$$\operatorname{span}(u_1, ..., u_k, v_1, ..., v_{n-k})$$

We can write the image of T as followed

$$\operatorname{Im}(T) = \operatorname{span}(T(u_1), ..., T(u_k), T(v_1), ..., T(v_{n-k})) = \operatorname{span}(T(v_1), ..., T(v_{n-k}))$$

We claim that $Tv_1, ..., Tv_{n-k}$ form a basis for $\operatorname{Im}(T)$. This proofs that $Tv_1, ..., Tv_{n-k}$ spans $\operatorname{Im}(T)$, so it remains to show that those vectors are linear indep.
Let $a_1, ..., a_{n-k} \in K$, and assume $\sum_{i=1}^{n-k} a_i T v_i = 0$.
Since T is linear

$$T\left(\sum_{i=1}^{n-k} a_i v_i\right) = \sum_{i=1}^{n-k} a_i T v_i = 0$$

$$\Rightarrow \sum_{i=1}^{n-k} a_i v_i \in \ker(T)$$

Note $u_1, ..., u_k$ is a basis for $\ker(T)$

$$\Rightarrow \exists b_1, ..., b_k \quad \text{such that } a_1 v_1 + ..., a_{n-k} v_{n-k} = b_1 u_1 + ... + b_k u_k.$$

$$\Rightarrow a_1 v_1 + ..., a_{n-k} v_{n-k} - b_1 u_1 - ... - b_k u_k = 0$$

But $u_1, ..., u_k, v_1, ..., v_{n-k}$ are linear independent (they form a basis)

$$\Rightarrow a_1 = ... = a_{n-k} = b_1 = ... = b_k = 0$$

This proves that

$$Tv_1, ..., Tv_{n-k}$$

form a basis for $\operatorname{Im}(T)$
Conclusion

$$\dim(T) = n - k = \dim(V) - \dim(\ker(T))$$

$$\dim(\ker(T)) + \dim(\operatorname{Im}(T)) = \dim(V)$$

□

2 Dual Maps

2.a Lemma(18.a.7)

The elemnts

$$v_1^\star, ..., v_n^\star$$

form a basis for $V^\star$
We denote this basis by

$$B^\star = (v_1^\star, ..., v_n^\star)$$

Moreover

$$\forall l \in V^\star, \quad l = \sum_{i=1}^{n} l(v_i) v_i^\star.$$

Proof.
Let $l \in V^\star$. We claim that

$$l = \sum_{i=1}^{n} l(v_i) v_i^\star$$

$$\Leftrightarrow l(v) = \sum_{i=1}^{n} l(v_i) v_i^\star(v)$$

Indeed, l and its representative sum are linear maps $V \to K$, so it's enough to check that in the second equation holds for $v = v_1, v = v_2, ..., v = v_n$, bc $v_1, ..., v_n$ form a basis for V.
Set $v = v_k$

$$\sum_{i=1}^{n} l(v_i) v_i^\star(v_k) = \sum_{i=1}^{n} l(v_i) \delta_{ik} = l(v_k).$$

The cleaim just proven shows that $v_1^\star, ..., v_n^\star$ span $V^\star$
since

$$\dim(V^\star) = \dim(V) = n,$$

it follows that $v_1^\star, ..., v_n^\star$ form a basis for $V^\star$.

□

2.b Lemma & def. (18.a.9)

Let V, W be vect. spaces oves K. Let

$$T : V \to W$$

be a linear map. Define a new map

$$T^\star : W^\star \to V^\star, \quad T^\star(l) := l \circ T \quad \forall l \in W^\star$$
$$(l : W \to K)$$

Then $T^\star$ is a linear map. It is called the dual map to T.
Proof.
Let $l \in W^\star$, i.e. $l : W \to K$ linear.
Clearly $l \circ T : V \to K$ is a linear map (b.c. both T and l are), so $T(l) := l \circ T \in V^\star$. This shows $T^\star : W^\star \to V^\star$ has its target indeed $V^\star$ as claimed.

We'll show now that $T^\star$ is a linear map.
Let

$$l \in W^\star, \quad \alpha \in K$$

We have that

$$\begin{aligned} T^{\star}(\alpha l) &\in V^{\star} \\ \Rightarrow T^{\star}(\alpha l)(v) &= ((\alpha l) \circ T)(v) \\ &= (\alpha l)(Tv) \\ &= \alpha l(T(v)) \\ &= \alpha(l \circ T)(v) \\ &= \alpha(T^{\star} l)(v) \end{aligned}$$

$$\Rightarrow T^{\star}(\alpha l) = \alpha T^{\star}(l)$$

Similarly one shows

$$\begin{aligned} T^{\star}(l_1 + l_2)(v) &= ((l_1 + l_2) \circ T)(v) \\ &= (l_1 \circ T + l_2 \circ T)(v) \\ &= (l_1 \circ T)(v) + (l_2 \circ T)(v) \\ &= T^{\star (l_1)}(v) + T^{\star}(l_2)(v) \end{aligned}$$

$$\Rightarrow T^{\star}(l_1 + l_2) = T^{\star}(l_1) + T^{\star}(l_2)$$

□

2.c Lemma (18.a.11)

Let

$$S : V \to W$$

be a linear map between two f.dim. Vect.spaces V & W over K. Let B be a basis for V and K a basis for w. Cosider

$$S^{\star} : W^{\star} \to V^{\star}$$

and the bases $\mathrm{K}^{\star}, \mathrm{B}^{\star}$ for $W^{\star}, V^{\star}$, respectibely. Then

$$[S^{\star}]_{\mathrm{B}^{\star}}^{\mathrm{K}^{\star}} = \left[S_{\mathrm{K}}^{\mathrm{B}}\right]^{T}$$

Proof.
Write

$$\begin{aligned} \mathrm{B} = (v_1, ..., v_n), &\quad \mathrm{K} = (w_1 ..., w_m) \\ A := [S]_{\mathrm{K}}^{\mathrm{B}}, &\quad A = (a_{ij}). \end{aligned}$$

A satisfies

$$S v_j = \sum_{i=1}^{n} a_{ij} w_i$$

Now for $w` \star \in \mathrm{K}^{\star}$ we have

$$\begin{aligned} S^{\star} w_j^{\star} &= w_j \circ S \\ &= \sum_{i=1}^{n} (w_j^{\star} \circ S)(v_i) v_i^{\star} \\ &= \sum_{i=1}^{n} w_j^{\star}(S(v_i)) v_i^{\star} \\ &= \sum_{i=1}^{n} w_j^{\star} \left(\sum_{k=1}^{n} a_{ki} w_k \right) v_i^{\star} = \sum_{i=1}^{n} a_{ji} v_i^{\star} \end{aligned}$$

$\Rightarrow$ coordinates of $S^\star w_j^\star$ in the basis $\mathrm{B}^\star = (v_1^\star, ..., v_n^\star)$ are

$$\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix}$$

row j in A, represented as col in the matrix A^T

$\square$

3 The iso theoreme 18.c.5

Let V, W be vect.sp.over K and

$$T : v \to W$$

a linear map. Define a new map

$$\overline{T} : V/_{\ker(T)} \to \mathrm{Im}(T,)$$

Let $x \in V/_{\ker(T)}$. Choose a represantative $v \in V$ for x.

$$x = [v]$$

Define

$$\overline{T}(x) := T(v)$$

Then

1. $\overline{T}$ well defined , and is linear map.
2. $\overline{T} : V/_{\ker(t)} \to \mathrm{Im}(T)$ is an isomorphism.
3. The following diagram cummutes

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "\pi"'] & W \\
V/\ker(T) \arrow[r, "\overline{T}"] & \mathrm{Im}(T) \arrow[u, "i"']
\end{tikzcd}$$

Proof.

3.a $\overline{T}$ is well defined.

Suppose $x = [v] = [v']$, we need to show that $T(v) = T(v')$.
Indeed $[v] = [v']$ implies that

$$\begin{aligned} v - v' &\in \ker(T). \\ &\Rightarrow T(v - v') = 0 \\ &\Rightarrow T(v) = T(v') \end{aligned}$$

3.b $\overline{T}$ is linear

$$\begin{aligned} \overline{T}(\alpha[v]) &= \overline{T}([\alpha v]) \\ &= T(\alpha v) \\ &= \alpha T(v) = \alpha \overline{T}([v]) \end{aligned}$$

$$\begin{aligned} \overline{T}([v_1] + [v_2]) &= \overline{T}([v_1 + v_2]) \\ &= T(v_1 + v_2) \\ &= T(v_1) + T(v_2) = T([v_1]) + T([v_2]) \end{aligned}$$

3.c $\overline{T}$ is injective

Let $x \in \ker\left(\overline{T}\right)$. Write x as $x = [v]$ for some $v \in V$.
Then

$$\overline{T}(x) = T(v) \Rightarrow v \in \ker(T) \Rightarrow x = [v] = 0_{v/_{\ker(T)}}$$

3.d $\overline{T}$ is surjective

Let

$$w \in \operatorname{Im}(T). \quad \Rightarrow \exists v \in V \text{ such that } T(v) = w$$
$$\Rightarrow \overline{[v]} = T(v) = w$$

This shows that

$$\operatorname{Im}(T) \subseteq \operatorname{Im}\left(\overline{T}\right).$$

But by the def. of $\overline{T}$ we also have $\operatorname{Im}\left(\overline{T}\right) \subseteq \operatorname{Im}(T)$

$$\Rightarrow \operatorname{Im}\left(\overline{T}\right) = \operatorname{Im}(T).$$

3.e The commutativity of the diagram

This follows immediately from the definitions.